

Savitribai Phule Pune University

Third Year of Electronics and Computer Engineering (2019 Course)

310341:Database Management System

Teaching Scheme:	Credit: 03	Examination Scheme:
Theory: 03 Hrs. / week		In-Sem (Theory): 30 Marks End-Sem (Theory): 70 Marks

Prerequisite Courses, if any: Data Structures and Algorithms

Companion Course, if any: Database Management System Lab

Course Objectives:

1. To understand fundamental concepts of database from its design to its implementation.
2. To analyze database requirements and determine the entities involved in the system.
3. To manipulate database using SQL Query to create, update and manage Database.
4. Be familiar with the basic issues of transaction processing and concurrency control.
5. To learn and understand Parallel Databases and its Architectures.
6. To learn and understand Distributed Databases and its applications.

Course Outcomes: On completion of the course, learner will be able to-

- CO1:** Understand the underlying concepts of database systems.
- CO2:** Design and implement a database schema for a given problem-domain using data model.
- CO3:** Solve wide range of query and update problems using SQL/DML/DDL commands.
- CO4:** Explain transaction Management in relational database System.
- CO5:** Understand various Database Architectures and its applications.
- CO6:** Apply NoSQL database concepts for processing unstructured data.

Course Contents

Unit I	Introduction to DBMS	(07 Hrs.)
Introduction to Database Management Systems, Purpose of Database Systems, Database-System Applications, Data Abstraction and Database System Structure.		
Relational Model: Structure of relational databases, Domains, Relations, Keys – Super key, Candidate key, Primary key, Foreign key, Relational algebra – fundamental operators and syntax, relational algebra queries, tuple relational calculus.		
Entity-Relationship model: Basic Concepts, Entity Set, Relationship Sets and Weak Entity Sets, Mapping Cardinalities, E-R diagrams, Extended E-R Features, Converting E-R & EER diagram into tables.		

Mapping of Course Outcomes for Unit I	CO1: Understand the underlying concepts of database systems.	
Unit II	Relational Database Design	(06 Hrs.)
Basic concepts, CODD's Rules, Relational Integrity: Domain, Referential Integrities, Enterprise Constraints, Database Design: Features of Good Relational Designs, Normalization, Atomic Domains and First Normal Form, Decomposition using Functional Dependencies, 2NF, 3NF, and BCNF.		
Mapping of Course Outcomes for Unit II	CO2: Design and implement a database schema for a given problem-domain using data model.	
Unit III	Basics of SQL	(07 Hrs.)
DDL, DML, DCL, Structure: Creation, Alteration, Defining constraints – Primary key, Foreign key, Unique key, Not null, Check, IN operator, Functions - Aggregate Functions, Built-in Functions –Numeric, Date, String Functions, Set operations, sub-queries, correlated sub queries, Use of group by, having, order by, join and its types, Exist, Any, All, view and its types. Transaction control commands: Commit, Rollback, Save-point PL/SQL Concepts: Cursors, Stored Procedures, Stored Function, Database Triggers		
Mapping of Course Outcomes for Unit III	CO3: Solve wide range of query and update problems using SQL/DML/DDL commands.	
Unit IV	Database Transactions Management	(07 Hrs.)
Basic concepts of a Transaction, Transaction Management, Properties of Transactions, Concept of Schedule, Serial Schedule, and Serializability: Conflict and View, Cascaded Aborts, Recoverable and Non-recoverable Schedules, Concurrency Control: Need, Locking Methods, Deadlock handling and Time-stamp based Protocols.		
Mapping of Course Outcomes for Unit IV	CO4: Explain transaction Management in relational database System.	
Unit V	Parallel and Distributed Databases	(06 Hrs.)
Multi-user DBMS Architectures, Introduction to Parallel Databases , performance measures-throughput, response time, speed-up, scale-up, Interconnection Network Architectures, Architectures for parallel databases, Evaluating Parallel Query in Parallel Databases, Virtualization on Multicore processors, Parallelizing individual operations, Parallel query optimizations. Introduction to distributed databases , Distributed DBMS architectures, storing data in a Distributed, Distributed Query processing, Updating distributed data, Distributed transactions, Distributed Concurrency control and Recovery.		
Mapping of Course Outcomes for Unit V	CO5: Understand various Database Architectures and its applications.	
Unit VI	No SQL Databases	(07 Hrs.)
Types of Data: structured, unstructured and semi structured data. NoSQL Database: Introduction, Need, Features. Types of NoSQL Databases: Key Value store, Document store,, graph, wide column store, BASE		

properties, DATA consistency model, ACID vs. BASE
MongoDB(with syntax and usage):CRUD operations, Indexing, Aggregation, Map Reduce, Replication, Shading.

Mapping of Course Outcomes for Unit VI

CO6: Apply NoSQL database concepts for processing unstructured data.

Learning Resources

Text Books:

1. A. Silberschatz, H.F. Korth and S. Sudarshan , “Database System Concepts”, McGraw Hill, 6th Edition.
2. C.J. Date, A. Kannan, S. Swamynathan “An introduction to Database Systems”, Pearson, 8th Edition.
3. Pramod Sadalage and Martin Fowler” NoSQL Distilled”, Addison-Wesley Professional

Reference Books:

1. Martin Gruber, “Understanding SQL”, Sybex Publications.
2. Ivan Bayross, “SQL-PL/SQL”, BPB Publications, 4th Edition.
3. S.K. Singh, “Database Systems: Concepts, Design and Application”, Pearson, Education, 2nd Edition.

MOOC / NPTEL Courses:

1. NPTEL Course “**Database Management System**”
Link of the Course: <https://nptel.ac.in/courses/106105175>
2. NPTEL Course “**Database Management System**”
<https://nptel.ac.in/courses/106104135>